2. From equation (1.29), the unit of electric potential is Joule per coulomb. The practical unit is volt (V) named after Alessandro Volta (1745-1827) who invented the electrical battery. The potential difference between two points is expressed in terms of volt.

Note

The description of motion of objects using the concept of potential or potential energy is simpler than that using the concept of field.

1.5.2 Electric potential due to a point charge

Consider a positive charge q kept fixed at the origin. Let P be a point at distance r from the charge q. This is shown in Figure 1.21.

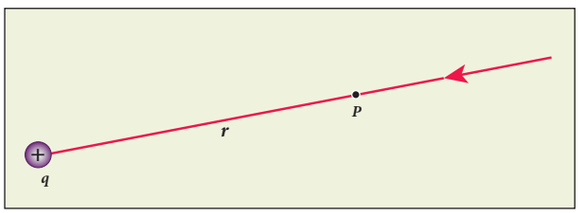


Figure 1.21 Electrostatic potential at a point P

The electric potential at the point P is

$$V = \int_{\infty}^{r} \left(-\vec{E}\right) \cdot d\vec{r} = -\int_{\infty}^{r} \vec{E} \cdot d\vec{r} \qquad (1.32)$$

Electric field due to positive point charge q is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$V = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^{r} \frac{q}{r^2} \hat{r} . d\vec{r}$$

The infinitesimal displacement vector, $d\vec{r} = dr\hat{r}$ and using $\hat{r} . \hat{r} = 1$, we have

$$V = -\frac{1}{4\pi\epsilon_0} \int_{\infty}^{r} \frac{q}{r^2} \hat{r} \cdot dr\hat{r} = -\frac{1}{4\pi\epsilon_0} \int_{\infty}^{r} \frac{q}{r^2} dr$$

After the integration,

$$V = -\frac{1}{4\pi\epsilon_0} q \left\{ -\frac{1}{r} \right\}_{\infty}^{r} = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Hence the electric potential due to a point charge q at a distance r is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \qquad (1.33)$$

Important points

(i) If the source charge q is positive, $V > 0$. If q is negative, then V is negative and equal to $V = -\frac{1}{4\pi\epsilon_0} \frac{q}{r}$

(ii) From expression (1.33), it is clear that the potential due to positive charge decreases as the distance increases, but for a negative charge the potential increases as the distance is increased. At infinity $(r = \infty)$ electrostatic potential is zero (V = 0).

In the case of gravitational force, mass moves from a point of higher gravitational potential to a point of lower

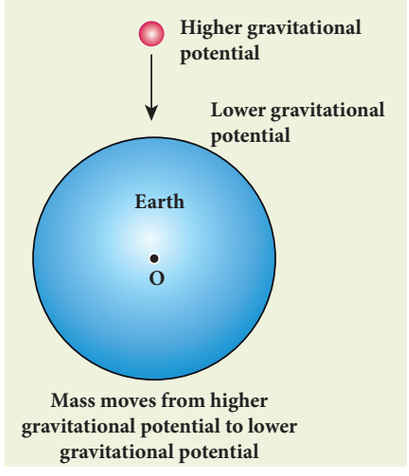


Figure 1.22 Motion of masses in terms of gravitational potential

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